Transient triglyceridemia decreases vascular reactivity in young, healthy men without risk factors for coronary heart disease.

BACKGROUND: Hypertriglyceridemia is now accepted as a risk factor for coronary heart disease, although the mechanism behind the increased risk is not well understood. The present study was undertaken to investigate the effects of triglyceridemia on endothelial function, because impaired endothelial function is considered a marker of atherogenesis. METHODS AND RESULTS: Flow- and nitroglycerin-induced dilatation of the brachial artery was investigated noninvasively by high-resolution ultrasound technique in seven young, healthy men without risk factors for coronary heart disease. Transient triglyceridemia was induced by infusion of a triglyceride emulsion, Intralipid, which raised free fatty acid concentrations twofold and triglyceride levels fourfold. Flow-induced vasodilatation decreased from 7.1+/-3.0% to 1.6+/-2.6% (P<.0002), whereas nitroglycerin-induced vasodilatation decreased from 20.5+/-5.8% to 11.5+/-3.2% (P<.002) before and after 1 hour of infusion of Intralipid, respectively. CONCLUSIONS: Transient triglyceridemia decreases vascular reactivity, presumably by both endothelium-dependent and endothelium-independent mechanisms.